Zheng Fang

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SUMMARY

Design engineer with five years in architecture and computational design and a Harvard GSD Master in Design (Technology track). I build interactive web products end to end: **Java / Spring Boot** and Python backend services, data pipelines, and visual, map-based interfaces in React and TypeScript.

EDUCATION

Harvard University **Cambridge, MA**
M.Des (Technology track), Cross-registered at MIT EECS *Sep. 2024 - Jun. 2026*
Coursework: Fundamentals of Programming, Data Visualization, Data Science I & II, ML Operations, Linear Algebra, Probability and Statistics.

Syracuse University **Syracuse, NY**
Bachelor of Architecture - Magna cum laude *Sep. 2017 - Jun. 2022*

TECHNICAL SKILLS

Java / Spring: Java, Spring Boot, Spring MVC (REST controllers), Spring Security, Spring Data JDBC, Java Servlet
Frontend: React, TypeScript/JavaScript, HTML & CSS, Redux, Vite, Mapbox GL JS, SVG, Ant Design
Other Backend: Python (FastAPI, Flask), Node.js/Express, Go, REST API design
Data and Cloud: PostgreSQL/MySQL, AWS (RDS, App Runner, Elastic Beanstalk, Lambda, EC2), GCP (GKE, Cloud SQL, Vertex AI), Docker, Kubernetes, GitHub Actions CI/CD
Computational Design and AI: Rhino + Grasshopper, computational geometry, GIS, image segmentation, LLM integration, RAG

EXPERIENCE

Harvard Graduate School of Design **Cambridge, MA**
Research Associate, with Rachel Meltzer *May 2025 - Sep. 2025*

- Studied how street-level transparency relates to perceived safety across Boston; the work became the first-author CAADRIA 2026 paper *Prompts to Perception*.
- Built a **Python + GIS** pipeline that sampled street points citywide and collected **49,740 Google Street View images**.
- Applied **promptable segmentation** to extract street features (building facade and window-to-wall ratio, sky, trees, road, sidewalk) as model inputs.
- Collected pairwise safety ratings on 300 images, trained models to predict perceived safety for the whole city, and explained them with **SHAP**; sky, trees, land use, road, buildings and sidewalks carried the most weight.
- Ran spatial statistics (Moran's I of 0.3 at 500 m, LISA clusters) and produced citywide safety and window-to-wall ratio maps.

PROJECT EXPERIENCES

TRACE: An Interactive Web Atlas of Acid Mine Drainage (React, Mapbox GL JS, Flask)

- Designed and built, solo and end to end, an interactive map that links coal mining operations to acid mine drainage discharges, the stream segments they affect, and the water-quality stations that document the harm. Submitted to the **GSD Digital Design Prize 2026**.
- Built the frontend in **React**, **Vite** and **Mapbox GL JS**: browse and analyze modes, ranked search lists, pollution heatmaps, 3D extrusion columns with a 2D/3D switch, and entity panels with hand-drawn **SVG** sparklines for monitoring chemistry.
- Animated a downstream particle simulation in the browser with **requestAnimationFrame**, showing pollution spreading along a 20 km stream chain from up to 30 sources at once.
- Wrote an offline **Python** pipeline that merges PA DEP, USGS NHDPlus (20,033 stream segments), USGS NWIS and EPA Water Quality Portal data (about 75K chemistry samples), served through a **Flask** API with four blueprints and deployed on **AWS Elastic Beanstalk**.

OnlineOrder: A Full-Stack Food Ordering System (Java, Spring Boot, PostgreSQL, React)

- Built the backend in **Java** with **Spring Boot**: REST endpoints in **Spring MVC** controllers for registration, menu search, cart, ordering and checkout.
- Structured the code as controller / service / repository layers wired with Spring **dependency injection**, so business logic stays separate from HTTP handling and database access.
- Modeled restaurants, menus, carts and orders in **PostgreSQL on AWS RDS**, accessed through **Spring Data JDBC**.
- Implemented session-based authentication with **Spring Security**, keeping the credential in an HttpOnly cookie instead of browser JavaScript and making logout and access revocation simple; chose this over stateless JWT, with a shared session store as the path to multiple instances.
- Built the **React + Ant Design** frontend for browsing menus, managing the cart and placing orders; containerized and deployed on **AWS App Runner**.

TopShot: An AI Photography Tutor with a RAG-Based Vision + Language Pipeline (team of 3)

- Co-built an AI photography tutor that analyzes uploaded photos and gives technical and composition feedback, grounded by retrieval over a curated dataset of about 25K photos.
- Owned the **retrieval backend**: image embedding and exemplar retrieval that grounds feedback generation on **Vertex AI**, served through **FastAPI**.
- Designed the **PostgreSQL (Cloud SQL)** database and built the data processing for the 25K-photo dataset, including **EXIF** metadata extraction.
- Shipped with **Docker Compose** to **GKE** via Artifact Registry, with **GitHub Actions** running ruff, pytest unit/integration/e2e tests, image builds and deploys on every push to main.

TopoFold: Transformable Structures from Topologically Complex Surfaces (Python, Rhino + Grasshopper)

- Designed and built, end to end, a method that turns topologically complex surfaces (Seifert surfaces) into deployable scissor-linkage lattices, a case earlier methods limited to simple ruled or radial surfaces could not handle.
- Wrote a **Python** bisection search along surface geodesics for point pairs whose normals intersect (0.001° tolerance), placing each hinge at the geodesic midpoint, then marched pairs into chains and assembled them into a quad grid.
- Handled unreliable normal intersections near high-curvature boundaries with a parallel-normal fallback, and avoided parity conflicts between strips with paired angulated linkages at each hub; the **Rhino + Grasshopper** pipeline outputs a closed, self-intersection-free, CAD-ready lattice.

RESEARCH AND PUBLICATIONS

- Fang, Z., Tan, Y., Hoberman, C.
  TopoFold: Transformable Structure Generation from Topologically Complex Surfaces via Normal Intersection Search.
  Proceedings of CAADRIA 2026, Vol. 1, pp. 367–376. Derives kinematically valid scissor linkages from Seifert surfaces while preserving global topology.
- Fang, Z., Li, X.
  Prompts to Perception: A Promptable Segmentation Approach to Transparency and Safety.
  Proceedings of CAADRIA 2026. Uses promptable segmentation on Boston street-view imagery to study how facade transparency shapes perceived safety.
- Fang, Z., Li, X., Sun, M., Tian, R.
  Sectional Sampling: Controlled Generation of High-Resolution Architecture Design with Diffusion Model and Latent Interpolation.
  ACADIA 2024. Generates high-resolution architectural designs with a diffusion model, using sectional sampling and latent interpolation for control.
- Anonymous Authors
  Vinema: A Tangible, Vinyl-Inspired Device for Human–AI Musical Co-Creation.
  Manuscript under review, 2025.